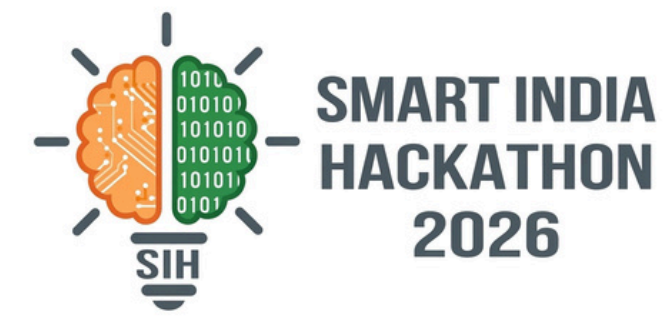
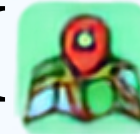
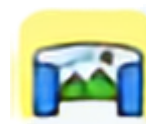


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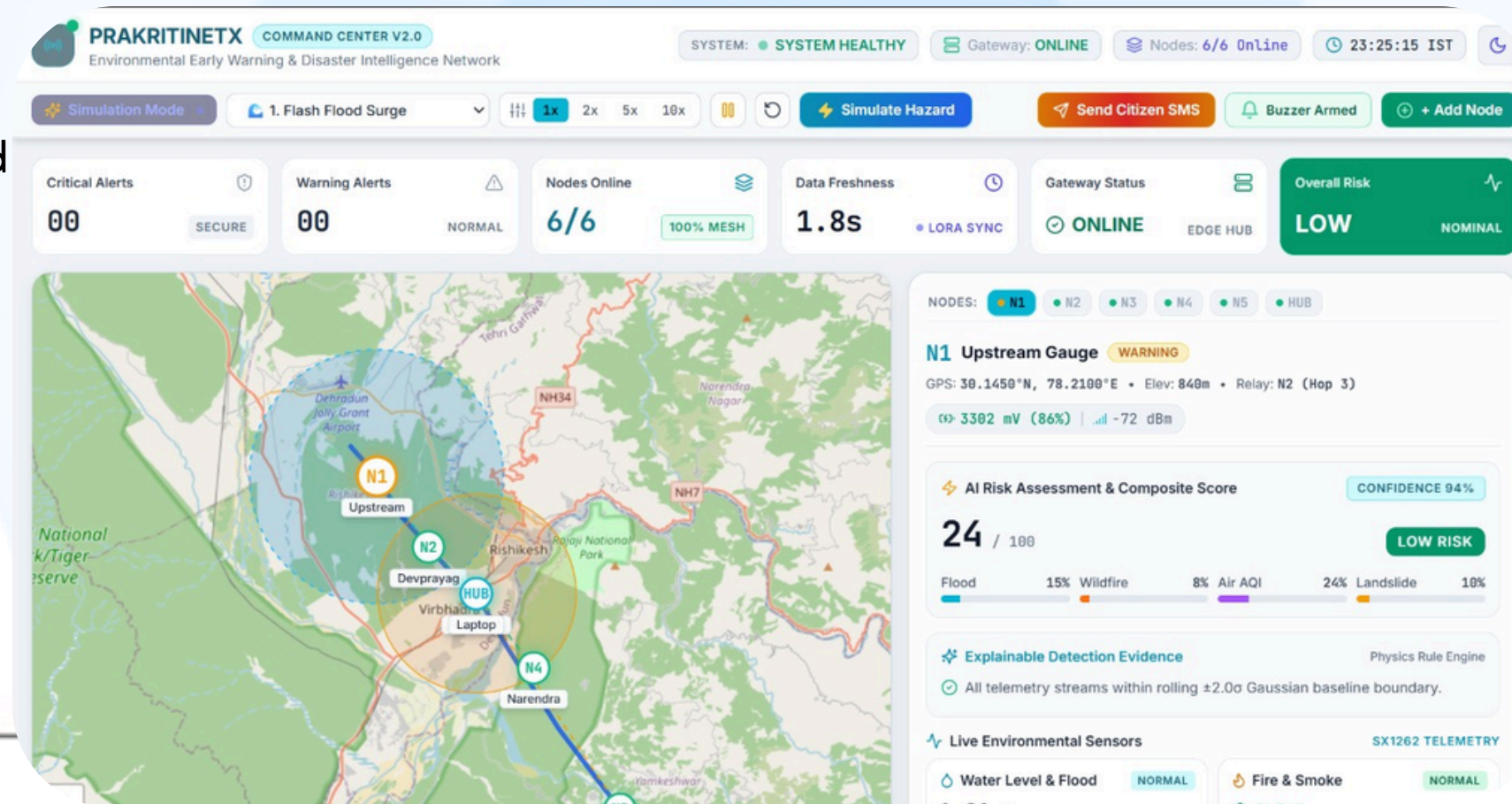
- Problem Statement ID - **SIH26178**
- Problem Statement Title - **Autonomous Flood & Fire Monitoring / Early Warning**
- Theme - **Disaster Management**
- PS Category - **Hardware**
- Team ID -
- Team Name (Registered on portal) - **PrakritiNetX**





Proposed Solution

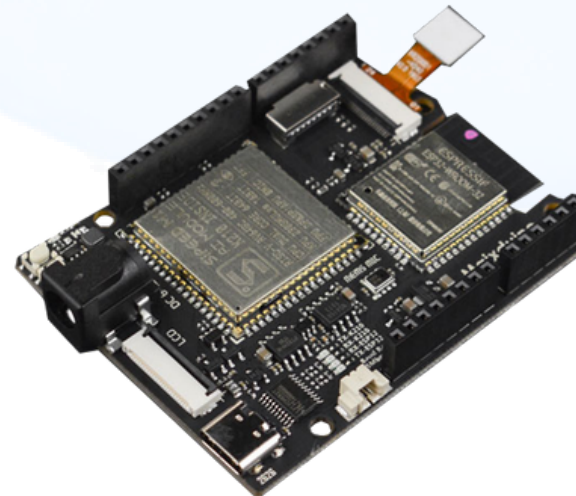
- **Multi-hazard sensing**
 - Flood level, temperature, smoke/fire and environmental sensing.
- **Local intelligence**
 - ESP32 classifies NORMAL, RISKY and share it via. LoRa Mesh Network
- **Resilient power**
 - Solar → MPPT → LiFePO4 → regulated power rails.



Android



IOS



It Addresses the Problem by

- **Continuous monitoring**
 - Reduces dependence on manual inspection.
- **Early warning**
 - LoRa sends alerts to a gateway over long range.
- **Disaster continuity**
 - Battery and energy priority keep critical sensing active.



INNOVATION & UNIQUENESS

Before • During • After

Monitoring, warning, response support and recovery monitoring.

Edge + network architecture

Decision happens at the node; communication stays long range.

Energy-aware operation

Critical loads stay ON; non-critical loads can be reduced/OFF.

Tech Stack

Edge AI & Hardware

Kendryte K210 ESP32-S3 0.5 T KPU
RISC-V 64b nncase 8-bit Quant

Multi-Hazard Sensors

OV5647 Telephoto Davis 7852 Rain
YG1006 IR Flame ADXL355 Low-Noise
SCA100T Dual-Tilt Capacitive Soil v1.2

Mesh & Connectivity

433 MHz LoRa SX1276 Radio P2P
Multi-Hop Mesh LoRaWAN SIM7600
UART / SPI 12C Bus 4G LTE Mod

Predictive Hub & Gateway

Raspberry Pi 4/5 SQLite DB Python
Scikit-Learn Z-Score 3σ 't = 'd / v

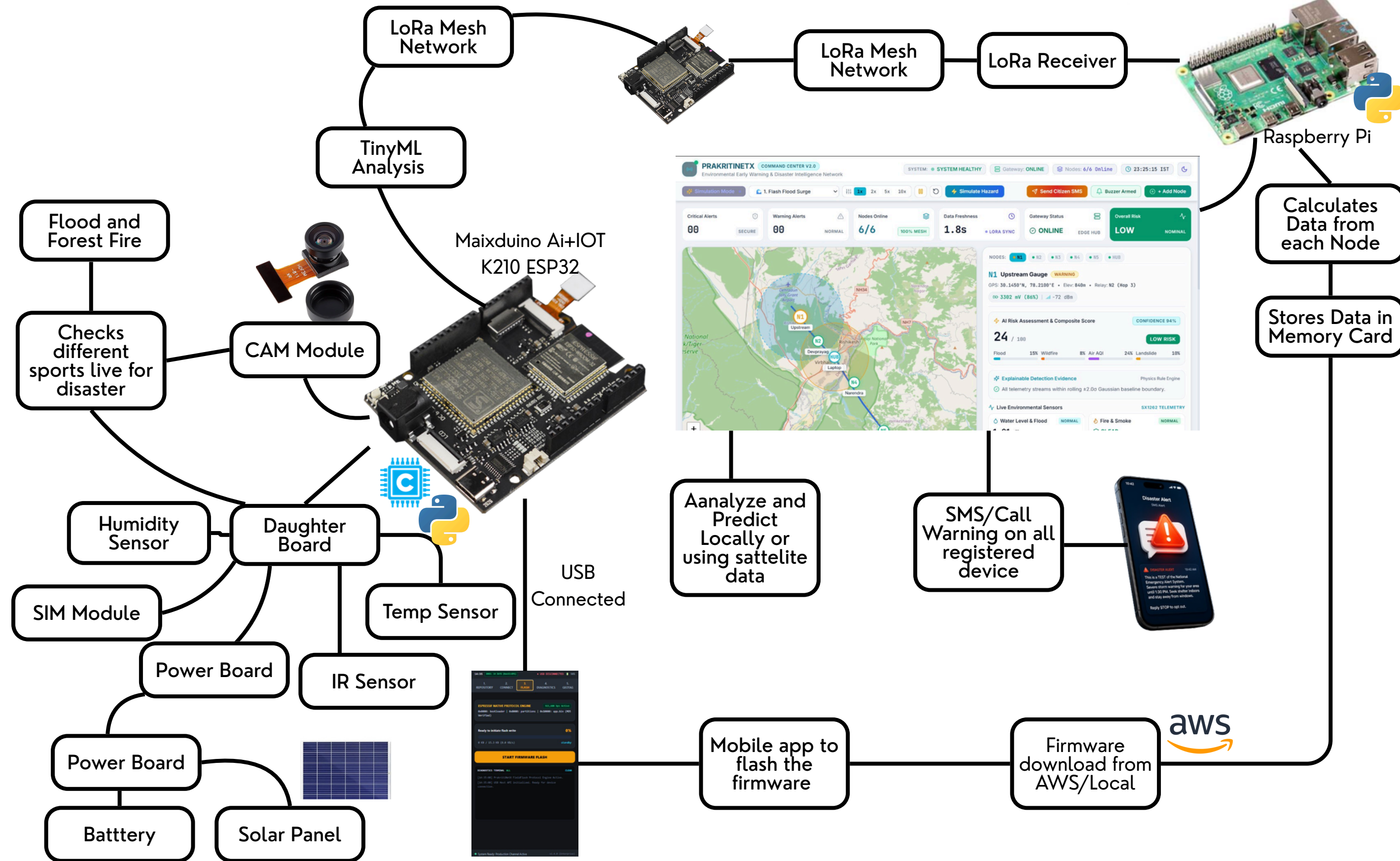
Alerting & Civil Defense

SIM800L GSM Direct SMS Twilio API
NDMA SACHET / CAP DistrictDMA Ready

Deployment & Mobile Tools

Android Kotlin USB-OTG GNSS Geotag
esptool / kflash React Vite Tailwind

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

Financials



Power Systems:

12V 15–20Ah LiFePO4 packs,
monocrystalline solar, MPPT chargers

Modular Daughterboards:
Rain gauges, flame sensors,
seismic accelerometers

Central Hub & Enclosures:
Raspberry Pi field gateway,
IP65 weatherproof housings

Vision & Compute Nodes:
Kendryte K210 + ESP32
modules, M12 telephoto lenses

Telemetry & Repeaters:
433 MHz LoRa transceivers,
antennas, GSM modules

Feasibility



Silicon Edge Inference: Dedicated KPU runs quantized MobileNet CNNs at 0.3W, eliminating cloud dependency.



Elevated Long-Range Optics: 20–30m elevated mount with narrow-FOV lenses survives 15–25m flood surges



Sub-GHz LoRa Mesh: 433 MHz radio penetrates mountain valleys and foliage via 14-byte compact binary packets.



Hardware Power-Gating: ESP32 timer cuts K210 idle draw to 0 mA, slashing consumption to ~150–180 mAh/day.

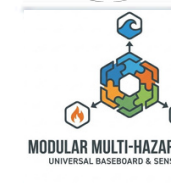
Viability



Audit-Ready Escalation: Transparent, rule-based logic complies with government disaster-management standards.



Zero-Calibration Adaptability: 30-day rolling baselines auto-adjust to any river geometry.



Modular Multi-Hazard Design: Universal baseboard supports floods, forest fires, and landslides via swappable sensor boards.



Actionable Lead Time: Upstream kinematic modeling provides hours of warning vs. minutes from water-level detectors.

Challenges and Mitigation

Extended Dark/
Monsoon Periods



10–14 day LiFePO4 battery autonomy + adaptive wake-interval throttling (10 to 60 mins).

Mesh Relay vs. Sleep
Conflicts



Split power domains: ESP32/LoRa stays listening (Domain A) while K210/camera sleeps (Domain B).

Mountain Line-of-
Sight Blockage



Multi-hop mesh networking with dedicated solar-powered repeaters placed along high ridgelines.

Mud, Algae & Lens
Obstruction



Downward-angled rain hoods + firmware-level contrast checks that auto-flag "sensor obstructed".

IMPACT & BENEFITS

PrakritiNetX

IMPACT



From Detection to Prediction

Moves beyond fixed-threshold detection by identifying abnormal trends before the hazard reaches downstream locations.



Multi-Node Flood Intelligence

Upstream and downstream nodes correlate changes to estimate hazard propagation and arrival time.



AI-Based Environmental Sensing

Edge AI cameras extract turbidity, turbulence and water coverage, while AI supports fire/smoke detection at the edge.



Resilient Remote Monitoring

Solar + battery powered nodes with LoRa mesh communication enable monitoring where cellular/Wi-Fi connectivity is unreliable.



BENEFITS



Communities

Receive zone-based SMS alerts when a propagating hazard threatens their area.



Disaster Management Authorities

Get explainable, auditable alerts based on sensor trends, rate of change and cross-node confirmation.



Remote Infrastructure

Solar-powered nodes with battery-health monitoring and adaptive power management improve long-term field reliability.



Researchers & System Operators

Continuous historical data helps improve site-specific baseline, propagation estimates and future prediction models.



Modular & Scalable

Swapable hazard-specific sensor boards support flood, fire and landslide monitoring using a common communication and power architecture.

Research Papers

[Github Link](#)

<https://github.com/Rohan-Gupta99/PrakritiNetX>

Flood/Disaster Early Warning Systems

- Remote Early Warning System for Mountain Floods with Robust ZigBee Wireless Networks – Mobile Networks and Applications, Springer, 2023
- Early Warning of Impending Flash Flood Based on AIoT – EURASIP Journal on Wireless Communications and Networking, 2022 (directly relevant: uses LoRa + SIM900 cellular backup for mountain slope monitoring, near-identical to your dual LoRa/SIM architecture)
- LoRa-based IoT Sensor Node for Real-time Flood Early Warning System – IOP Conf. Series: Earth and Environmental Science, 2022
- Flood Early Warning System by Twitter Using LoRa – river-level ultrasonic sensing + Raspberry Pi processing + mesh topology, ResearchGate
- Development of a Smart Sensing Unit for LoRaWAN-based IoT Flood Monitoring and Warning System in Catchment Areas – ScienceDirect, 2023

Edge AI / On-device Vision

- We designed and implemented WaterNet, a lightweight custom CNN architecture (approximately 90K parameters) for on-device river water segmentation, deployed directly on the K210's hardware KPU via the nncase v0.2 compilation pipeline. This is an original contribution of our project rather than an adaptation of prior published work – full architecture, training pipeline, and export process are documented in our repository (Software/ML_Model/) and technical reference document.

Government Applications

- NDMA Common Alerting Protocol (CAP) and SACHET platform – National Disaster Management Authority, Government of India
- Central Water Commission (CWC) flood forecasting operational standards – Government of India